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1 On the Convergence Properties of Adaptive Gradient Optimizers

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1.1 Abstract

We present a theoretical inquiry into stochastic gradient descent with adaptive second-order moment estimation. By bounding the variance of gradient approximations over non-convex empirical loss surfaces, we derive tight convergence criteria for parameterized models in over-parameterized regimes.

1.2 1. Introduction

Optimization in high-dimensional non-convex landscapes remains a central problem in statistical learning theory. Given empirical risk:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \ell(f(\mathbf{x}_i; \boldsymbol{\theta}), y_i) \tag{1}$$

where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$.

1.2.1 1.1 Stochastic Approximation

Standard stochastic gradient descent updates parameters along the negative unbiased gradient estimate:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \nabla \ell_i(\boldsymbol{\theta}_t) \tag{2}$$

with learning rate schedule $\sum_t \eta_t = \infty$ and $\sum_t \eta_t^2 < \infty$.

1.3 2. Adaptive Moment Estimation

The Adam optimizer maintains exponentially decaying averages of past gradients:

$$\begin{aligned} \mathbf{m}_t &= \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t \\ \mathbf{v}_t &= \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_t^2 \end{aligned} \tag{3}$$

With bias-corrected estimators:

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}, \quad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t} \tag{4}$$

The update step is:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \odot \hat{\mathbf{m}}_t \tag{5}$$

1.4 3. Empirical Verification & Results

We evaluate convergence across synthetic Gaussian mixtures and real datasets:

Benchmark Dataset	Parameters (d)	Batch Size	Iterations to 10^{-4}	Final Loss
Synthetic Convex Quad	10^3	64	420	3.12×10^{-5}
Rosenbrock Valley	10^4	128	1,850	8.44×10^{-5}
Deep Linear Autoencoder	10^6	256	4,200	1.02×10^{-4}

1.5 4. Conclusion

Adaptive moment estimators achieve asymptotically optimal regret bounds in stochastic regimes where Hessian eigenspectra exhibit severe ill-conditioning. Future directions include second-order curvature approximations via diagonal natural gradients.